Vision and Need

FreeMoCap [1] is a free and open-source markerless motion capture software system that produces research-grade full body kinematic data from generic cameras (Figure 1). Built and maintained by the FreeMoCap Foundation since 2021, it has grown organically into a vibrant, successful project with nearly 10,000 GitHub stars, telemetry indicating more than 15,000 users across 153 countries, and a global community forum of over 4,000 students, researchers, educators, and artists.

The project's broad adoption and continued growth reveal a real and unmet need. Quantitative movement measurement supports work across biomechanics, neuroscience, rehabilitation, and animation, yet conventional systems remain costly and are often tied to specialized hardware or fixed workflows. Researchers whose needs fall outside those workflows therefore continue to rebuild substantial portions of motion-capture infrastructure independently. By running on consumer-grade cameras and following established patterns for the development and management of free-and-open-source-software (FOSS), FreeMoCap provides a validated, research-grade measurement tool appropriate for high-level research laboratories as well as classrooms, garage labs, and basement mocap studios [2].

FreeMoCap now sits at a critical inflection point in its growth curve. As is common for mid-scale FOSS projects, our user count has grown far faster than its maintainer base, a community pattern that Nadia Eghbal's study of open source software, *Working In Public*, refers to as a "Stadium" [3] - many users relying on the output of a proportionally tiny team of developers. If growth in the user base continues to outpace its maintainers, the growing weight of support runs the risk of overwhelming the core team. Sustaining the project through this period requires building the organizational and community infrastructure that supports a transition toward what Eghbal calls a "Federation," a pattern representing a healthy balance of user and contributor growth, allowing technical knowledge and responsibility to be distributed across the community.

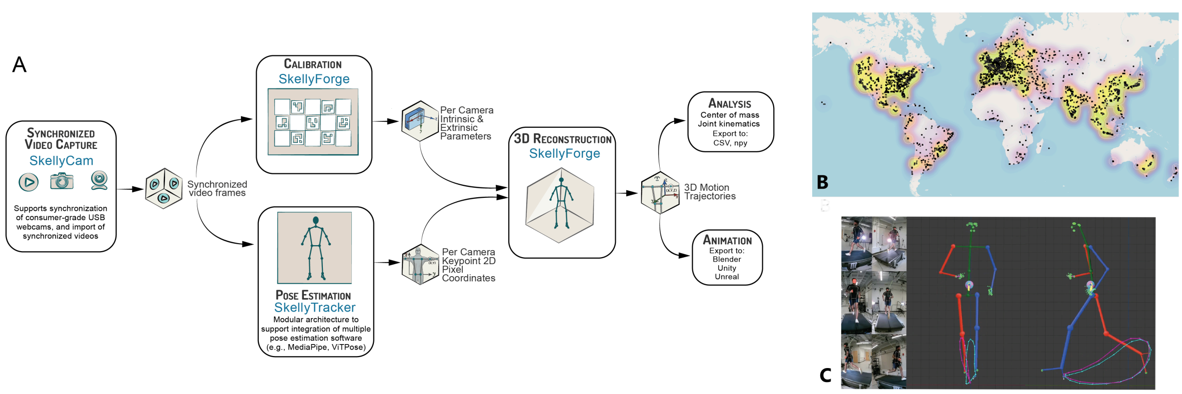

Figure 1: (A) The FreeMoCap workflow and its component repositories (blue). (B) Global FreeMoCap usage. (C) Reconstructed 3D trajectories alongside the source camera views.

Despite this growth, our understanding of the broader FreeMoCap community remains limited. Telemetry and interactions with users who participate in public forums do not reliably tell us who our users are, what they need, or where existing technical and community infrastructure create barriers.

This proposal requests Track 1 support to scope the foundation for future growth: understanding our **community landscape** and its needs; developing **documentation and upskilling pathways** that convert user interest into contributor capacity; governance practices that create **transparency and productive communication** between the core maintainers and the wider community; a **validation and benchmarking methodology** that other laboratories can replicate; and a **sustainability model** for the maintenance that the software will require.

Current State of the Software and Community

The FreeMoCap codebase is hosted in a collection of public GitHub repositories under the `freemocap` organization (`github-dot-com-slash-freemocap`), built and tested through automated CI/CD pipelines, and distributed as a cross-platform OS-certified standalone desktop application via installers hosted through our website (`freemocap-dot-org-slash-download`).

The project is organized as a polyrepo - the core functionality of the application is split across domain-scoped constituent sub-repositories — `skellycam` for camera synchronization and recording, `skellytracker` as a modular interface for pose estimation, `skellyforge` for calibration, 3D reconstruction, and kinematic analysis, and `freemocap_blender_addon` for export to Blender (and other animation software) (Figure 1). Each sub-repository represents an independently functional codebase and follows the same internal layout, so that documentation, tests, and entry points sit in the same place in every repository. This self-similar structure makes the full codebase substantially easier to teach, to learn, and to navigate with AI coding assistants, and a contributor with expertise in one domain can work inside a single component without holding the whole system in mind.

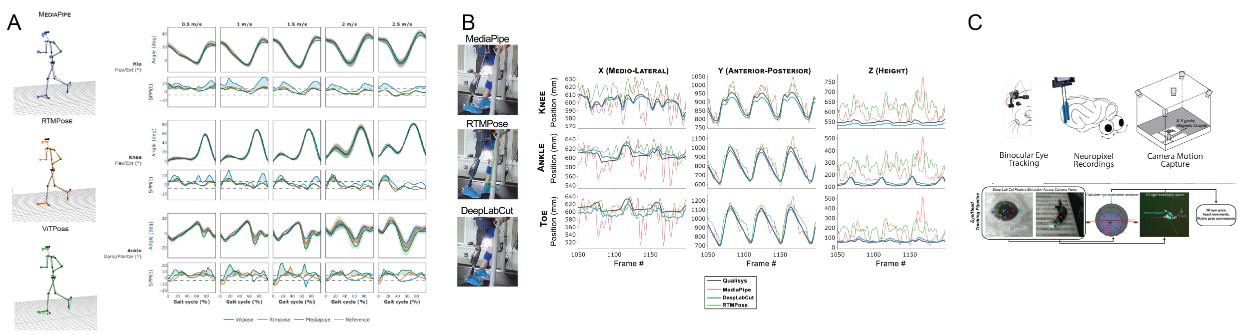

Figure 2: FreeMoCap applications from standard human biomechanics to specialized movement-analysis use cases. A. Lower-extremity joint-angle comparison and statistical parametric mapping against Qualisys marker-based motion capture across multiple pose-estimation backends. B. Custom DeepLabCut integration for tracking a lower-limb prosthesis user. C. Extension to ferret locomotion and neural-behavioral research in collaboration with the Scholl Lab, CU Denver Anschutz

FreeMoCap is used across several distinct communities, including athletes, clinicians, educators and students, technical artists, and general technologists. Published use of FreeMoCap spans reaching and grasping biomechanics [4], gait analysis [5], computational linguistics [6–8], animation [9], and robotic exoskeleton design and verification [10]. Independent validation studies have reported agreement with marker-based motion capture across several movement tasks [11, 12], while our own work [2] has characterized gait and balance performance, demonstrated integration of multiple pose-estimation backends, and extended the workflow to non-human locomotion (Figure 2).

Despite growing external contributions, much of the architectural and operational knowledge required for core development remains concentrated within a core development team, and pathways for contributor onboarding and remain immature. Track 1 scoping will ground the development of these mechanisms in evidence about the community they are intended to serve.

Team Qualifications

The FreeMoCap team represents decades of individual scientific and engineering expertise, with more than five years of direct experience developing the FreeMoCap software specifically, including supporting its community, and managing the technical and organizational challenges associated with its growth.

Ecosystem Discovery

Need for Innovation

A variety of motion capture tools currently exist, but each is built around particular assumptions about users, domains, hardware, or deployment. Commercial systems include biomechanics-focused platforms such as Theia3D and animation tools such as Move.ai and Rokoko, while open-source alternatives such as OpenCap [13], Pose2Sim [14], and PosePipe [15] are largely centered on applications of human biomechanics. These systems also impose different operational constraints: Theia3D relies on expensive hardware, while OpenCap's standard workflow requires supported iOS devices, account log-in screens, and remote cloud processing.

Researchers with specialized requirements still build substantial portions of this pipeline independently, as evidenced by the continued development of domain-specific workflows in individual laboratories [16–19], often reimplementing the stages shown in Figure 1. Across projects, this duplicates technical effort and reduces the return on public research investment in software that is difficult to maintain, reproduce, or transfer beyond the environment in which it was developed.

Our Track 1 scoping will test the premise that the **repeated rebuilding of motion-capture pipelines is an ecosystem failure**, as researchers looking for extensible tools often find building from scratch easier than extending shared infrastructure. What is needed is not only accessible motion-capture software, but an ecosystem in which specialized adaptations developed by one community can be contributed back, maintained, and reused by others — infrastructure that depends on defined stewardship and governance as much as on code. Track 1 will determine what documentation, contribution pathways, technical interfaces, and governance structures are needed to reduce the barriers to extending and contributing to shared infrastructure

Necessity of an open-source ecosystem approach

FreeMoCap's existing community provides early evidence that an open-source ecosystem can convert individual expertise into shared infrastructure for the broader user community. Users have already become contributors in areas including animation, UI design, and GPU-accelerated pose-estimation architecture, bringing domain expertise from outside the core development team into the project for the benefit of all users. However, these transitions occurred informally and have depended heavily on direct interaction with the core team. Creating a repeatable path into the ecosystem first requires understanding **who** is trying to use, teach, extend, or contribute to FreeMoCap, **what** technical knowledge each group needs, and **where** the current software and documentation fail to support them independently.

Stakeholder discovery

Our current understanding of the FreeMoCap community is drawn primarily from users who contact us directly or participate in existing community channels. Some applications have emerged outside the domains we originally anticipated. For example, computational-linguistics researchers have adopted FreeMoCap in published work [6–8] without any prior interaction with the core team. This motivates a broader discovery effort aimed not only at *known users*, but also at **understanding unexpected communities**, what drew them to the platform, and what would reduce barriers to sustained and broader adoption.

Our proposed work will combine analysis of existing community activity, community-wide surveys, and the I-Corps for PESOSE stakeholder-discovery process to characterize four user populations reflecting the communities already observed around FreeMoCap: (1) **researchers** using FreeMoCap as part of scientific projects, (2) **educators** building FreeMoCap-based teaching materials, (3) **technical artists** working in 3D animation and related fields, and (4) current and prospective **contributors**. Discovery will produce three assessments, each addressing one condition under which extending shared infrastructure becomes easier than rebuilding it.

1) **Cross-domain needs assessment**: how different communities use FreeMoCap, what brought them to it, and what their work requires that our standard pipelines do not yet provide.

1) **Knowledge assessment**: what users must understand to *operate* the software, what contributors must understand to *modify* it, which of that knowledge remains undocumented, and what forms of documentation or instruction would allow people to participate without direct access to the core team.

2) **Comparative ecosystem review**: how mature open-source projects of comparable or larger scale structure contribution, extension, and decision-making, and how those structures perform as projects grow. These findings will supply candidate models for evaluation in *Organization and Governance* and inform the security, community-building, and sustainability work that follows.

Organization and Governance

Governance Scoping

Healthy open source ecosystems employ methods to distribute decision-making and knowledge such that it does not depend on any one person. In the Python Enhancement Proposal (PEP) process, upcoming software changes are announced and discussed via public Request For Comments (RFCs) that invite community input but (critically) do not gate progress on strict consensus. NumPy and SciPy run comparable processes (NEP and SEP, respectively) for stakeholder-facing transparency. What these share is a way of keeping consequential decisions visible and open to comment without making every decision wait on consensus. Track 1 will determine how these patterns should be adapted for a project of FreeMoCap's scale, shaped by the community landscape revealed through ecosystem discovery.

Distributed infrastructure and community plug-ins

FreeMoCap's modular architecture provides a technical basis for community plug-ins. Because the component repositories are organized around fixed domains — cameras, calibration, image analysis, reconstruction — rather than around any particular model or library, a new pose-estimation method can be adopted without disturbing the rest of the pipeline. For example, the `skellytracker` architecture exposes an open interface through which AI-based pose-estimation models developed by different groups can interoperate with a shared scientific workflow without modification to the underlying pipeline (Figure 2 A). This interface matters because no single pose-estimation model is adequate across all subjects. The algorithms in common use are trained on standard human datasets, and the requirements of biomechanical measurement differ from those the models were built to satisfy [20].

A **community plug-in**, in this context, is code written and maintained by someone outside the core team that runs against a published interface without altering the shared pipeline. A researcher whose subjects fall outside existing models could implement a tracker against a documented interface rather than assembling an entire pipeline around it, and the result would be available to everyone facing the same problem, while an AI-tracking method published by a computer-vision researcher can be translated into a reusable scientific workflow for users who would otherwise lack the technical expertise to implement it.

A community plug-in system is a classic protection against "scope creep" - the ecosystem can grow in capability without the core team absorbing the maintenance of every addition, and users can adopt new capabilities without waiting for the core maintainers to implement them. Extensions that prove useful additions through community use can be pulled into the core. Community plug-ins allow for a separation between the development of the core architecture (on available to high level developers and trusted contributors), and softer, lower-stakes context of plug-in development.

Track 1 will define the review, distribution, maintenance, and responsibility structures required for this Community Plug-In model, informed by projects already operating plug-in ecosystems and the needs and preferences of the members of our community who are likely to participate.

Licensing

FreeMoCap and its component repositories are licensed under the AGPLv3+. Because so much of the landscape of closed-source motion capture operates behind proprietary servers, the AGPLv3+ is the only standard license whose terms would stop a closed-source competitor from running `FreeMoCap` behind a server endpoint while charging users for access. Because the Foundation holds the rights to the codebase, it can also offer alternative terms to organizations whose intended use is incompatible with the AGPLv3+, keeping the software freely available to everyone else while generating resources for maintenance and stewardship.

Long-term Sustainability

The FreeMoCap Foundation was incorporated as an IRS-recognized 501(c)(3) public charity to provide an institutional home for the project. In parallel, the Foundation will evaluate a diversified model for the long-term sustainability of this research infrastructure, including alternative commercial licensing, standardized hardware kits geared towards students, with higher-end software targeting more resourced labs, training and support services, institutional partnerships, and continued grant funding. SkellyTechnologies, LLC, a separate for-profit entity associated with the project, provides an additional pathway for commercial services, SBIR/STTR proposals, B2B agreements, and other profit-seeking activities that may support the long-term sustainability of the ecosystem.

Risk Analysis and Security Plan

Video-based motion capture creates inherent privacy concerns because identifiable recordings must be captured and processed. These concerns can themselves become a barrier to adoption: recent work on OpenCap found that patients remained concerned about video sharing, storage, and access despite HIPAA-compliant cloud safeguards [21]. FreeMoCap avoids this cloud-data surface by processing video and derived data **locally**, without requiring a login or network connection, allowing sensitive recordings to remain under the user's direct control.

Secure and trustworthy research infrastructure also requires addressing **software-supply-chain** risk. FreeMoCap depends on a substantial stack of computer-vision, machine-learning, and scientific-computing libraries, and the proposed community plug-in system would introduce additional third-party code into the processing environment. Track 1 will assess FreeMoCap against security standards maintained by the Open Source Security Foundation (OpenSSF), which provides established criteria and certification pathways for secure open-source development. This assessment will identify gaps and the requirements needed to reach an appropriate certification level, informing both a project security roadmap and the review, approval, and release controls needed to guard against malicious or inadvertently unsafe community plug-ins.

Data Integrity
While the fidelity of FreeMoCap-produced data has been and continues to be validated, **preserving the integrity of that data (i.e., that modifications to the software do not silently alter the measurements it produces) is an ecosystem responsibility**. Track 1 will scope stage-specific integrity checks across the processing pipeline, with expected outputs tracked across software versions. Camera characteristics are also part of the measurement chain. Collaboration with Imatest LLC, whose work includes ISO/TC 42 standards activity on image-information content, will inform how imaging-system limitations can be characterized, reported, and made visible to users.

Because these are the same checks an outside laboratory would need in order to trust the software on its own data, Track 1 will also scope how we should package and communicate our validation and benchmarking methodology, so that other groups can reproduce our validation on their own hardware and contribute the results back as community validation.

Identity and Telemetry
Potential ecosystem features such as contributor recognition, authenticated accounts, and usage telemetry introduce additional privacy and security considerations. Track 1 will scope approaches that minimize collected data and preserve FreeMoCap's local-first principles, including evaluation of established authorization approaches such as OAuth 2.0 for (optional) account-based features. The project's industry mentor, Endurance Idehen, will advise this work, including authentication, authorization, telemetry, and related security controls. Together, these activities will produce a risk register, draft extension review and release controls, and a data-integrity and security roadmap.

Community Building

The FreeMoCap Foundation maintains an active Discord community forum of more than 4,000 members, structured so that users can ask and answer questions without relying exclusively on the core development team. The core team also hosts a weekly community call for project updates, user feedback, technical discussion, and interaction with potential contributors. These activities provide an established foundation for community engagement, but **rapid user growth in 2026 has increased the need for education and participation models that can scale beyond direct interaction with maintainers.**

Track 1 will develop and pilot accessible educational and participation pathways at three levels: **introductory materials** for general users; **intermediate and domain-specific resources** for scientific, technical, creative, and educational applications; and separate **developer documentation** covering software architecture, contribution workflows, testing, and maintenance.

These materials will be piloted through hands-on classes and workshops with partner organizations, including The Possible Zone and Artisans Asylum, and through tutorials and guided projects made available to the broader community. Pilots will identify where learners encounter conceptual or technical barriers, which formats work best for different audiences, and what knowledge can be transferred without direct core-team instruction.

Track 1 will also test low-barrier forms of community participation that do not require software development, including community-created visualizations, artwork, examples, challenges, and other ways for users to contribute domain expertise.

Together, these activities will inform a scalable education and onboarding model that supports progression from user, to informed participant, to sustained contributor. Longer term, the resulting model could inform a larger FreeMoCap community meeting or conference bringing together users, developers, educators, and researchers as part of a subsequent Track 2 effort.

Milestones and Evaluation Plan

Evaluation

Track 1 success will be evaluated by: (1) documented governance roles, contributor pathways, and software architecture, assessed by whether contributors outside the core team can act on them without direct consultation; (2) educational and onboarding materials, assessed through pilot participant performance and observed barriers; (3) a framework for developing, reviewing, distributing, and maintaining community plug-ins, assessed through limited pilots of the proposed plug-in development framework; (4) a documented validation and benchmarking methodology, assessed by whether an independent laboratory can reproduce our accuracy checks on its own hardware and recordings; and (5) a sustainability model, assessed by review of its revenue and maintenance assumptions against the demand evidence gathered during discovery. Each output will be revised in response to stakeholder feedback before incorporation into the final ecosystem plan.

Qtr	Milestone / Deliverable	Evaluation
Q1–Q2	Complete I-Corps for PESOSE; establish baseline ecosystem metrics	Completion confirmed by I-Corps instructors
Q1–Q2	Stakeholder discovery; prioritized user and contributor barrier map	Target groups covered; recurring barriers confirmed
Q2	Develop candidate governance, contributor, and stewardship models	Compared with discovery findings and reviewed by OSE maintainers
Q2–Q3	Define extension contribution, maintenance, licensing, and review requirements	Tested against discovered extension needs; external OSE review
Q3	Security/data-quality scoping; prototype benchmarking and regression suite	Known output differences detected; risks mapped to controls
Q3	Pilot education and onboarding with partner organizations	Participant feedback and observed barriers
Q4	Integrate findings into governance, sustainability, community, and implementation roadmap	Processes usable without direct core-team consultation

Broader Impacts

In a fully developed ecosystem, the same software can serve a thirteen-year-old with a couple of webcams and a federally funded research laboratory studying human perceptuomotor neuroscience. A contribution made by an animator can improve the software for researchers and clinicians, and scientific advancements can become accessible to artists, educators, students, and hobbyists. A sustainable FreeMoCap ecosystem can turn specialized adaptations into shared research infrastructure, increasing the return on public investment while accelerating the translation of research advances across fields and into practice.

The educational activities piloted under this proposal will help develop and test materials for users of differing technical backgrounds to progress towards independent software use, and where desired, contribution. A mature ecosystem can teach a user how a pixel in an image becomes a 3D point in space, not just how to run the software that does it. By pairing extensible infrastructure with education and cross-domain participation, FreeMoCap, supported by a developed ecosystem, can broaden not only who has access to quantitative measurement tools, but also who is able to understand, adapt, and help shape them.